# EE150 Signal and System
# Homework 3

Note:

- Please provide enough calculation process to get full marks.
- Please submit your homework to Gradescope in PDF version.
- It's highly recommended to write every exercise on a single sheet of page.
- Late submissions will have points deducted according to the penalty policy.
- Please use English only to complete the assignment, solutions in Chinese are deducted according to the penalty policy.
- Plagiarizer will get zero points.
- The full score of this assignment is 100 points.

## Exercise 1. (10pt)

The impulse response of a LTI system is

$$h(t) = e^{-5(t-2)}u(t-2)$$

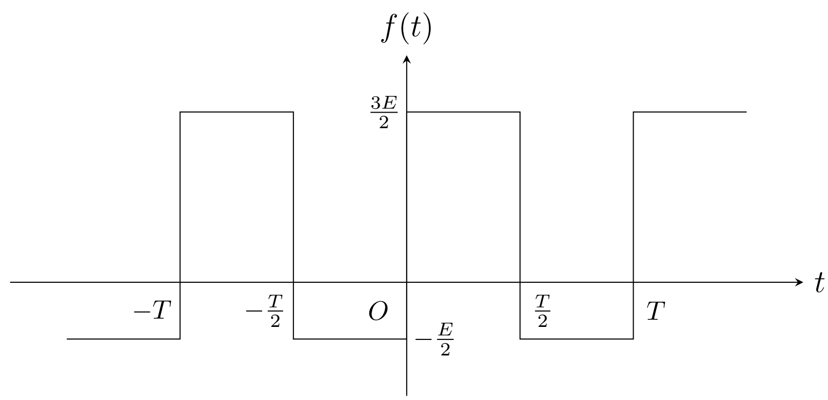
Calculate the following:
(a) $H(s)$
(b) The output signal $y(t)$ of the system when the input signal is $x(t) = 3\sin(2t)$.

## Exercise 2. (20pt)

(a) Find the fundamental frequency $\omega_0$ and the non-zero Fourier series coefficients $a_k$ for the following signal:

$$x(t) = 9 + 8\cos\left(\frac{2\pi}{5}t\right) + 6\sin\left(\frac{3\pi}{4}t\right) + 4\cos\left(\frac{7\pi}{10}t\right)$$

(b) Find the Fourier series coefficients of the **periodic** signals shown in the following figure.



## Exercise 3. (15pt)

(a) Given that the Fourier series coefficients of a periodic signal $x(t)$ with fundamental period $T = 4$ are

$$a_k = \begin{cases} jk + |k|, & |k| < 3 \\ 0, & else \end{cases}$$

find the **trigonometric form** of the original signal $x(t)$.
(b) Let $x_1(t)$ be a continuous-time periodic signal with fundamental frequency $\omega_1$ and Fourier series coefficients $a_k$. It is known that $x_2(t) = 3x_1(5-t) + 4x_1(t-3)$. What is the relationship between the fundamental frequency $\omega_2$ of $x_2(t)$ and $\omega_1$? And find the relationship between the Fourier series coefficients $b_k$ of $x_2(t)$ and the coefficients $a_k$.

## Exercise 4. (10pt)

Given three continuous-time periodic signals with fundamental period $T = \frac{1}{3}$

$$x(t) = \sin(6\pi t), \quad y(t) = \cos(6\pi t), \quad z(t) = x(t)y(t)$$

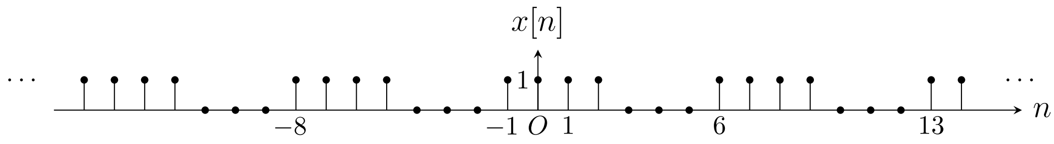
(a) Find the Fourier series coefficients of $x(t)$ and $y(t)$, then use the multiplication property of the Fourier series to find the Fourier series coefficients of $z(t)$.
(b) Calculate the average power $P$ of $\frac{dz(t)}{dt}$ using differentiation properties of Fourier series and Parseval's theorem.

# Exercise 5. (10pt)

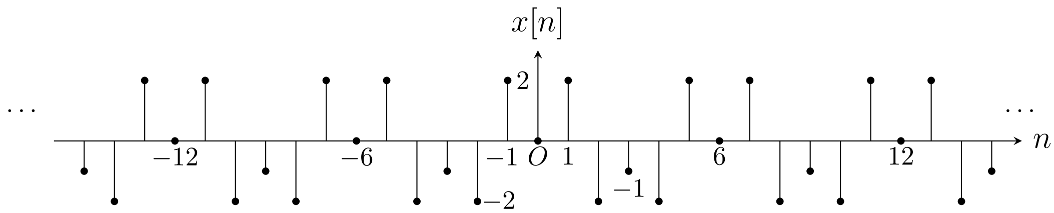
Regarding a discrete-time periodic signal $x[n]$ with Fourier series coefficients $a_k$, the following information is given:
(1) $x[n]$ is a real signal
(2) $x[n]$ has a period of $N = 4$
(3) Average power $P = 6$
(4) $a_0 = 1, \quad a_{17} = 1 + j, \quad a_6 \in \mathbf{R}$ and $a_6 > 0$
Try to determine the original signal $x[n]$.

## Exercise 6. (15pt)

Find the Fourier series coefficients for each of the following discrete-time periodic signals.




$(a)$




$(b)$ Tips: $x[0] = 0$

## Exercise 7. (20pt)

Given the input periodic signal and LTI system impulse response, find the frequency response and output signal .

(a) $x(t) = \sin(3t) + 2\cos(4t), \quad h(t) = e^{-3|t|}$

(b) $x[n] = \cos(\frac{2\pi}{3}n) + 2\cos(\frac{4\pi}{5}n), \quad h[n] = e^{-3n}u(n)$